

Solution Architecture

Team ID	SWUID2026383049
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Team Size	4
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The solution follows a three-layer architecture: Presentation, Application and AI layer.

Layer	Components	Responsibility
Presentation Layer	index.html, style.css, JavaScript	Collects user input, shows answers, quiz and roadmap
Application Layer	main.py (FastAPI), qna.py, explanation_module.py, summary_module.py, quiz_module.py, learning_path.py	Routing, validation, prompt building and response formatting
AI Layer	gemini_helper.py, Google Gemini 1.5 Pro/Flash, LaMini-Flan-T5	Generates content, retries and falls back across models
Configuration	Environment file (.env)	Stores GEMINI_API_KEY securely

API Endpoints

Endpoint	Method	Input	Description
/	GET	-	Renders the web interface
/qa	GET / POST	question	Concise answer
/explain	GET / POST	topic	Simple explanation
/summarize	POST	text	Summary of passage
/quiz	POST	text	3 MCQs in JSON
/learn/recommendations	GET / POST	topic	Learning roadmap
/api/config	GET / POST	api_key	Check or update API key